

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

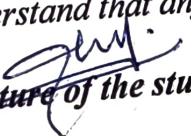
Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I K.Thanuja ¹⁹²³⁷³⁰¹² (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formulas [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient age data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

Define Data mining and list any two basic data mining tasks. explain the steps involved in the KDD process and justify the role of data preprocessing in it.

Definition

Data mining is the process of extracting useful patterns, relationships, trends and knowledge from large amount of data.

Two Types:

1. classification
2. clustering

Steps in the KDD

1. Data cleaning : Remove missing values, noise, and inconsistencies.
2. Data Integration: combines data from multiple sources into one dataset.
3. Data selection: select only relevant data for mining.
4. Data Transformation: converts data into suitable forms.
5. Data mining: Applies algorithm to discover pattern.
6. Pattern evaluation: Identify interesting and useful patterns.
7. Knowledge presentation: Displays result using graph.

2. What is Data cleaning? Mention two common data quality issues. explain methods to handle missing value and noisy data with suitable ex:

Data cleaning is the process of correcting inaccurate, incomplete, duplicates or inconsistent data.

Two data quality issues

1. Missing values
2. Noisy data

Handling missing values

1. Ignore the tuple : used when only a few records are missing.

ex :

Age	Salary
25	50000
?	45000

Ignore second record

2. Replace missing values

ex :

$$\text{Average Age} : (22 + 24 + 26 + 28) = 25$$

Replace missing value ⁴ with 25.

Handling Noisy Data

1. Binning : smoothing values mean or median
2. Regression : fits a line and predicts correct values.
3. Define Data Integration and list two challenges associated with it. Explain how redundancy and inconsistency.

Defination : Data Integration combines data from multiple sources into one unified dataset.

Challenges

1. Redundancy
2. Data inconsistency

Handling Redundancy

ex:

Database A
Student ID

101

Database B

Roll NO

101

Both represent the same student.

remove duplicates records.

use correlation analysis

Handling Inconsistency

convert both into one standard format

Apply data transformation value.

4. What is Covariance Analysis?

Covariance measures whether two variables increase or decrease together.

Population

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

sample

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

5. Suppose two stocks A & B having following values.

A = 2, 3, 4, 5, 6 B = 5, 8, 10, 11, 14 mean of A = 4, mean of B = 9.6

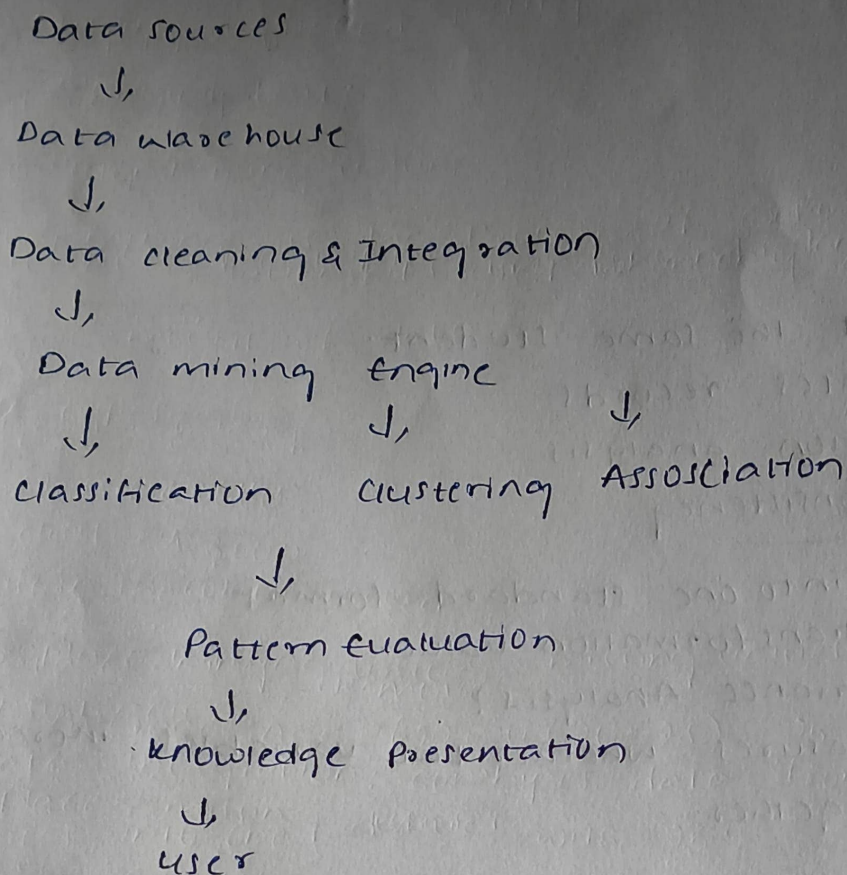
A	B	(A-4)	(B-9.6)	Product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

Total = 20

$$\frac{20}{5} = 4$$

(Positive)

6. explain the architecture of a Data mining system



7.

covariance.

Given $x = (4, 5, 6, 7, 8)$

$y = (8, 3, 7, 5, 6)$

Mean of x

$$\bar{x} = 6.2$$

mean of y

$$\bar{y} = 5.8$$

x	y	$(x - 5.8)$	Product $(x - 5.8)$	$(x - 5.8)$ Product
4	8	-2.2	-11.84	-2.2
5	3	-2.8	3.36	-1.2
6	7	0.2	-0.24	-0.2
7	5	-0.8	-0.64	0.8
8	6	0.2	0.56	2.8

$$\text{sum} = -1.8$$

Population covariance

$$= \frac{-1.8}{5} = -0.36$$

(Negative)

8.

Describe the major steps in Data processing

1. Data cleaning
2. Data Integration
3. Data Transformation
4. Data reduction
5. Data Discretization

9.

Perform equal-frequency (equi-depth) binning

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 22, 25, 25, 25, 30, 33, 33, 33, 35,
35, 35, 35, 36, 40, 45, 46, 52, 70

27 values

9 values

Bin 1

13, 15, 16, 16, 19, 20, 20, 21, 22

mean = 18

smoothed bin

18 18 18 18 18 18 18 18

Bin 2

25, 25, 25, 25, 25, 25, 25, 25, 25, 4

Bin 3

28, 28, 28, 28, 28, 28, 28, 28, 28, 28

$$= \frac{394}{9} = 44$$

Bin 3

44, 44, 44, 44, 44, 44, 44, 44, 44, 44

10. Data Discretization converts continuous values into discrete interval (bins)

Formula

$$\text{Width} = \frac{\text{Max} - \text{Min}}{\text{No. of bins}}$$

ex: 10, 20, 30, 40, 50

3 bins

$$\text{Width} = \frac{50 - 10}{3} = 13.33$$

Bins

= =

10 - 23.33

23.34 - 36.66

36.67 - 50

equal - frequency binning

Data: 10, 20, 30, 40, 50, 60

Bin 1: 10, 20

Bin 2: 30, 40

Bin 3: 50, 60

comparison

Feature

equal-width

equal-frequency

Interval size

equal

unequal

Records per bin

varies

equal

Outliers

less

Best for

~~unequal~~
yes
uniform data

skewed data.